

Substance Use & Misuse

ISSN: 1082-6084 (Print) 1532-2491 (Online) Journal homepage: www.tandfonline.com/journals/isum20

A Descriptive Assessment of Products Containing the Opioid Receptor Stimulator Mitragynine Pseudoindoxyl

C. Michael White, Jordyn Belcourt & Ava Sedensky

To cite this article: C. Michael White, Jordyn Belcourt & Ava Sedensky (2025) A Descriptive Assessment of Products Containing the Opioid Receptor Stimulator Mitragynine Pseudoindoxyl, Substance Use & Misuse, 60:12, 1950-1954, DOI: 10.1080/10826084.2025.2522162

To link to this article: https://doi.org/10.1080/10826084.2025.2522162

View supplementary material

Published online: 26 Jun 2025.

Submit your article to this journal

Article views: 308

View related articles

View Crossmark data

Citing articles: 8 View citing articles

Full Terms & Conditions of access and use can be found at
https://www.tandfonline.com/action/journalInformation?journalCode=isum20

SUBSTANCE USE & MISUSE
2025, VOL. 60, NO. 12, 1950–1954
https://doi.org/10.1080/10826084.2025.2522162

Taylor & Francis
Taylor & Francis Group

NOTE

Check for updates

A Descriptive Assessment of Products Containing the Opioid Receptor Stimulator Mitragynine Pseudoindoxyl

C. Michael White, Jordyn Belcourt and Ava Sedensky

University of Connecticut School of Pharmacy, Storrs, Connecticut, USA

ABSTRACT

Background: Potent semi-synthetic alkaloids that stimulate opioid receptors are being sold to consumers in gas stations, smoke shops, and online as the dietary supplement 'kratom'. *Objective*: We descriptively evaluated the characteristics of products containing mitragynine pseudoindoxyl. *Results*: Fifty-one unique products met our inclusion criteria. Seventy-one percent of products contained mitragynine pseudoindoxyl plus 7-hydroxymitragynine, 24% contained mitragynine pseudoindoxyl alone (24%), and 6% contained mitragynine pseudoindoxyl with 8-hydroxymitragynine or 11-hydroxymitragynine (with or without 7-hydroxymitragynine). Sixty-nine percent of products had a child appealing flavoring and/or scent, 63% of products were formulated using bright colors or had bright colors on their labeling, 12% of products had food pictures on the label (mainly representing the fruit or mint flavoring/scent), 6% of products had a cartoon mascot, and 6% of products had a food formulation (ice cream cone or chocolate bar). Only 18% of websites disclosed the products stimulated opioid receptors while 45% disclosed they could cause intoxication, tolerance, dependence, withdrawal, or addiction. *Conclusions*: There are many products that contain mitragynine pseudoindoxyl sold as dietary supplements and many have features that would appeal to children.

KEYWORDS
Kratom; mitragynine pseudoindoxyl; 7-hydroxymitragynine; opioid

Introduction

Fresh and dried leaves of the Mitragyna speciosa tree in Southeast Asia have been used as an herbal product called whole leaf kratom for energy boosting, relaxation, pain control, and opioid substitution for centuries (McCurdy et al., 2024). Mitragynine comprises as little as 12% and up to 66% of the total alkaloids in fresh Mitragyna speciosa leaves (whole leaf kratom) and primarily stimulates serotonin, alpha-1, and alpha-2 receptors while only indirectly and mildly stimulating mu-opioid receptors (Ponglux et al., 1994; Takayama et al., 1998). There are many other natural alkaloids in whole leaf kratom that reinforce the serotonin and alpha-2 agonist effects. When dried, some of the mitragynine (<2% of the total alkaloid content) degrades into the alkaloid 7-hydroxymitragynine (McCurdy et al., 2024).

Within the past two years, a proliferation of high dose semi-synthetic 7-hydroxymitragynine and mitragynine pseudoindoxyl products has occurred (White, 2024a). Preclinical studies show that these alkaloids potently stimulate mu-opioid receptors, are antagonized by opioid antagonists like naloxone and naloxonazine, and do not stimulate alpha-1 and alpha-2 adrenoceptors or serotonin receptors (Kruegel et al., 2019; Váradi et al., 2016; Yamamoto et al., 1999). It was previously shown in a rodent study that mitragynine use could mildly suppress respiration but had a ceiling effect preventing serious respiratory depression while 7-hydroxymitragynine suppressed respiration and did not have such a ceiling effect (Hill et al., 2022). A recent rodent study found that 7-hydroxymitragynine significantly suppressed rodent breathing frequency, tidal volume, and minute volume at higher doses, while mitragynine did not (Zuarth-Gonzalez et al., 2025). The impact of mitragynine pseudoindoxyl on respiratory depression was assessed in a rodent model and respiration was found to be unaffected at a low dose but reduced by 15% at a higher dose (Váradi et al., 2016). A recent letter to the editor from kratom researchers discussed that 7-hydroxymitragynine is a risky ingredient in products available to consumers without a prescription because of concerning animal studies showing physical dependence and withdrawal effects with a lack of human safety and efficacy trials (Smith et al., 2025). The same is true for mitragynine pseudoindoxyl as demonstrated by the development of physical dependence and withdrawal signs and symptoms in morphine addiction rodent models as evidenced by increased diarrhea, jumping, and wet shake frequency occurring when naloxone was administered or when treatment with this alkaloid was tapered (Wilson et al., 2021).

We sought to determine how many products for oral administration are sold over the internet that contain mitragynine pseudoindoxyl, what other alkaloids they contain, their serving sizes, whether they could be attractive to children, and what safeguards are available to prevent children from using them.

CONTACT C. Michael White Charles.white@uconn.edu University of Connecticut School of Pharmacy, 69 North Eagleville Rd, Storrs, CT06269, USA.
Supplemental data for this article is available online at https://doi.org/10.1080/10826084.2025.2522162
© 2025 Taylor & Francis Group, LLC

Materials and methods

We performed a systematic search in the Google search engine using the terms "mitragynine pseudoindoxyl" or "Red-OH" (a pseudonym of mitragynine pseudoindoxyl) from January 10th to March 1st, 2025. Google was the only engine searched because it is used in over 88% of searches in the United States, and to reduce the identification of the same available products across multiple platforms (Oberlo, 2024). We did not use a VPN and conducted the search in the United States. Since the study was not on animal or human participants or their personal or medical data, it is exempt from IRB review.

We included all orally available products sold online that contained mitragynine pseudoindoxyl with or without other alkaloid ingredients. We found 21 distinct retailers selling at least one product containing mitragynine pseudoindoxyl alone or in combination with other alkaloids (Supplemental Table S1). We determined if the websites made shoppers check a box specifying they were over 18 or 21 years to enter and whether they required age verification before purchase.

We recorded all products meeting our criteria from each site. However, if a website had duplicative products that we discovered from a prior website, we only included the original website's data unless there was missing data that a subsequent website included. We recorded the alkaloids they contained, the suggested serving size, their enteral product formulations, whether the packaging was considered child-resistant (a child-resistant cap on a vial or bottle), whether the product labeling specified age restrictions to purchase, and the features of the product that would appeal to children. Since blister packing of over the counter or prescription drugs meets the criteria for being child resistant, we collected this information although we could not assure that the products did not have easy to open features.

We searched all the websites we identified and the labeling for each of the unique products they contained for the following terms: opioid, intoxication, impairment, dependence, tolerance, withdrawal, addiction, or habit-forming.

Table 1. Product formulations.

Active ingredient	Products found, No. (% of 51 total products)
Mitragynine pseudoindoxyl alone	
Chewable tablet	7
Liquid	5
Total	12 (23%)
Mitragynine pseudoindoxyl and 7-hydroxymitragynine	
Chewable tablet	30
Liquid	4
Ice cream cone	1
Chocolate bar	1
Total	36 (71%)
Mitragynine pseudoindoxyl with 8-hydroxymitragynine or 11-hydroxymitragynine (±7-hydroxymitragynine)	
Chewable tablet	2
Ice cream cone	1
Total	3 (6%)

No.: number.

Results

Supplemental Tables S2–S4 provide a synthesis of our raw data (Amsterdam Smoke, n.d.; CBD American Shaman, n.d.; Delta 8 Resellers, n.d.; Gala, n.d.; Great CBD Shop, n.d.; OPMS, n.d.; Pure Leaf Kratom, n.d.; The Powder Solutions, n.d.). Overall, we found 51 unique products that met our inclusion criteria (Amsterdam Smoke, n.d.; CBD American Shaman, n.d.; Delta 8 Resellers, n.d.; Gala, n.d.; Great CBD Shop, n.d.; OPMS, n.d.; Pure Leaf Kratom, n.d.; The Powder Solutions, n.d.). The largest group of products contained mitragynine pseudoindoxyl with 7-hydroxymitragynine (71%), followed by products with mitragynine pseudoindoxyl alone (24%), and then products with mitragynine pseudoindoxyl with 8-hydroxymitragynine or 11-hydroxymitragynine (with or without 7-hydroxymitragynine) (6%) (Table 1). Seventy-six percent of these products were chewable tablets, 18% were liquids, 4% looked like dried ice cream cones with ice cream, and 2% were chocolate.

Only one product (2%) came in a bottle with a child-resistant cap. Thirty-four products (67% of the total) came in a blister pack which may or may not be child resistant. The child resistant packaging from 4 products (8%) could not be determined, and 12 products (24%) were not child-resistant because they were: (1) in bottles with easy-open twist-off caps, (2) they were just wrapped or in an easy open box, (3) or because the label stated they were "not child-resistant." Overall, 37 products (73%) had an age restriction warning specified on their packaging, but 14 products (27%) did not.

We observed several product features that would appeal to children (Table 2). Sixty-nine percent of products had an appealing flavoring/scent [berry (raspberry, strawberry, wild berry), mint, watermelon, pink lemonade, candy apple, grape, citrus, mango, pistachio, and vanilla bean], 63% of product packaging or the tablets or liquid itself were formulated using bright colors, 12% of products had food pictures on their packaging (usually representing the flavoring/scent), 6% of products had a cartoon mascot, and 6% of products looked like an ice cream cone or chocolate bar. The company names include allusions to opioids including "Opia" for opioids, "Perks" for Percocet, "HiJoy" for the opioid euphoric effect, and "Lucid." Some child enticing product names included "Blue Magic Riz," "Purple Ease," "Purple Rein," "Purple Hornets," "Green Giants," "Yellow Jackets," and "The OG." While two of the eight retail websites (25%) used an honor code question of whether the searcher was over the age of 18 or 21 years old to enter, the rest (75%) did not. No site required proof of age to purchase a product.

Table 2. Product features that appeal to children.

Product feature	Products, No. (% of the 51 total products)
Flavoring/scent	35 (69%)
Mascot	3 (6%)
Formulation resembles food (i.e. ice cream cone, chocolate)	3 (6%)
Label features image of food	6 (12%)
Formulation is a bright color*	32 (63%)

No.: number.
*Excludes white, black, gray, tan, and brown.

Table 3. Website and/or label key words.

Opioid key word	Products, No. (% of 51 total products)
Opioid agonist or binds to opioid receptors	5 (9)
Produces "opioid-like effects"	2 (3)
Interacts with brain receptors like opioids	3 (6)
Other key words	**Products, No. (% of 51 total products)**
Intoxication	4 (8)
Impairment	4 (8)
Dependence	10 (20)
Tolerance	5 (10)
Withdrawal	5 (10)
Addiction	9 (18)
Habit-forming	3 (6)

Several products required splitting of a tablet or liquid to provide the suggested serving size. The tablet serving size may have been a full tablet, ½ tablet, or ¼ tablet (with the tablet not being scored for ¼ tablet splitting). Liquid serving sizes ranges from ½ bottles to a sixth of the bottle (with no liquid dosing aid to precisely measure the amount). Both ice cream cones products required splitting, either into halves or thirds with no scoring on the product to demarcate a serving. The chocolate bar listed a proper serving size as ½ a square.

We searched the website descriptions and the product labeling or packaging for the 51 unique products that they sold for our prespecified key terms with results summarized in Table 3. None of the product labels or packaging had the term "opioid" but 18% of the product descriptions on the websites did. Overall, 9% of the websites described that the products bind to opioid receptors, 3% said the products had opioid-like effects, and 6% said the products interact with brain receptors like opioids. Forty-five percent of products had at least one warning on the website product descriptions or the product labels. Specifically, 20% of products used the term dependence, 18% addiction, 10% tolerance, 10% withdrawal, 8% intoxication, 8% impairment, and 6% habit forming. The impairment term was followed by instructions not to operate heavy machinery.

Discussion

This is the first descriptive evaluation of products that contain the opioid stimulating semi-synthetic alkaloid mitragynine pseudoindoxyl with or without 7-, 8-, 11-hydroxymitragynine to our knowledge (McCurdy et al., 2024; White, 2024a; Wilson et al., 2021). These products are being sold over the internet as well as in gas stations, smoke shops, and convenience stores across the United States. However, given the semi-synthetic alkaloids that they contain, they do not meet the criteria to be a dietary supplement and are experimental drugs (McCurdy et al., 2024). FDA requirements for dietary supplements require companies to identify that they are new dietary ingredients if they were not in general use before 1994 and then wait until the required timeframe for FDA review before they can be sold (FDA, 2024a). That was not done with mitragynine pseudoindoxyl products before they were sold and mitragynine pseudoindoxyl is not an alkaloid found in fresh Mitragyna speciosa leaves (McCurdy et al., 2024). Dietary supplements also cannot make actual health claims unless an authoritative body has identified sufficient evidence to substantiate it, and the seller alerts the FDA that they plan to make a health claim and wait for the required timeframe for the FDA to review the claim (FDA, 2024b). These products are being touted for real health claims, such as pain relief without FDA notification. Tianeptine, which is a tricyclic antidepressant with potent opioid receptor stimulation, and these alkaloids have been termed "gas station heroin" because of their broad accessibility and ability to induce euphoria and opioid-like use disorders (Francis, 2024; White, 2024b). There are important parallels between tianeptine and the mitragynine pseudoindoxyl alkaloid products (Smith et al., 2025). Namely, the ability to access products in unlimited quantities with no medical oversight that potently stimulate mu-opioid receptors with data suggesting that they can cause physical dependence and withdrawal symptoms (Kruegel et al., 2019; McCurdy et al., 2024; Takayama et al., 1998; Váradi et al., 2016; Yamamoto et al., 1999).

We found that many of these products would appeal to children given their flavorings and scents, ice cream cone or chocolate bar formulations, cartoon mascots, and coloring of their tablets and/or packaging (Global Kratom Coalition, 2024; White, 2024b). Only one product (2%) clearly met the criteria for being child resistant. While most of the products are in blister packs which may or may not be child resistant, 23% of products are clearly not child resistant and could lead to inadvertent exposure if young children find them in a home. Although 73% specified that the products were not for use by children, young children may not be able to read the packaging. The child-friendly formulations and packaging may also entice teens (Soneji et al., 2019). Children with access to a credit card can purchase all these products with no age verification over the internet.

Some states ban "kratom" which includes bans on semi-synthetic alkaloids like 7-, 8-, or 11-hydroxymitragynine or mitragynine pseudoindoxyl. Nine states have specific Kratom Consumer Protection Laws that allow the sale of kratom products but specifically ban semi-synthetic 7-hydroxymitragynine or mitragynine pseudoindoxyl products and a few other states are currently considering such legislation (Global Kratom Coalition, 2025). In the 45 states without "kratom" bans, only Utah would consider an ice cream cone formulation of "kratom" to be illegal (Heflin, 2023). However, it is illegal in 14 states to sell cannabis products with this formulation. Furthermore, 13 states have limitations on images and words that can appear on cannabis product packaging to avoid appealing to children but have no such limitations for mitragynine pseudoindoxyl products (The Network for Public Health Law, 2022). The Federal Trade Commission and Food and Drug Administration have sent cease and desist letters to cannabis companies selling products that look like snacks or candy, but no letters have been sent to companies

selling mitragynine pseudoindoxyl products (Federal Trade Commission, 2023).

Many of the product serving sizes are not intuitive. The standard serving size ranges from ¼ to a whole tablet, two to six servings per bottle, two servings per chocolate square, and two to three servings per ice cream cone (Gala, n.d.; Great CBD Shop, n.d.; Pure Leaf Kratom, n.d.). The product suggesting ¼ tablet servings was not scored to allow precise splitting, the six serving liquid formulation did not contain a spoon or oral syringe dosing aid, and there were no instructions on how to split the asymmetric cone three ways (Gala, n.d.; Great CBD Shop, n.d.; Pure Leaf Kratom, n.d.). People not reading the packing carefully or children with limited reading comprehension may be prone to overconsumption.

Only 18% of product websites alluded to the opioid agonist nature of the products (McCurdy et al., 2024). Less than half of products alluded to the possibility of dependence, habit formation, addiction, tolerance, withdrawal, or intoxication. Mitragynine pseudoindoxyl and 7-hydroxymitragynine solely rely on opioid agonism for their pharmacologic effects and tolerance and addiction are important potential risks that should be disclosed (McCurdy et al., 2024).

Our assessment has some limitations. We were limited to the information that the online retailers placed on their websites. We did not visit brick and mortar stores to view products or purchase products to physically test them. By only searching the Google search engine we might have missed some products identified on other search engines or retail stores. Blister packs for over the counter and prescription products meet the criteria for being child-resistant but we did not physically purchase the mitragynine pseudoindoxyl products or test them to assure they did not have any easy open features or others that could render them not child resistant.

Conclusion

Products containing the semi-synthetic opioid receptor stimulating alkaloid mitragynine pseudoindoxyl, with or without 7-, 8-, or 11-hydroxymitragynine, are readily available through online retailers, even though they are not dietary supplements or approved drugs. Several of those products are designed in ways that may be alluring to small children and teens and do not have child resistant packaging. There are no firm restrictions for children with a credit card to purchase these products online. The serving sizes are not intuitive, risking an up to 4- to 6-fold increase in dosage by taking one tablet or the contents from an entire shot sized bottle. Since products with these alkaloids may cause physical dependence and withdrawal, preventing children from accessing these products is important and those risks should be disclosed but frequently are not.

Acknowledgments

None.

Ethical approval

Not applicable.

Declaration of interest

C. Michael White, Pharm.D. is the inaugural chair of the nonprofit Kratom Consumer Advisory Council. He provides pharmacology and health policy expertise to the nonprofit group of kratom consumers who have either been harmed or helped by kratom products and develop position statements about protecting consumer safety and product transparency.

Funding

This project is unfunded.

ORCID

C. Michael White http://orcid.org/0000-0002-9367-4893

Data availability statement

All source data is provided in Supplementary Tables.

References

Amsterdam Smoke (n.d.). Retrieved February 10, 2025, from https://amsterdamsmoke.shop/

CBD American Shaman (n.d.). Retrieved February 10, 2025, from https://cbdamericanshaman.com/

Delta 8 Resellers (n.d.). Retrieved February 10, 2025, from https://delta8resellers.com/

FDA (2024a, April 3). New dietary ingredient NDI notification process. Retrieved May 28, 2025, from https://www.fda.gov/food/dietary-supplements/new-dietary-ingredient-ndi-notification-process

FDA (2024b, March 28). Label claims for food and dietary supplements. Retrieved May 28, 2025, from https://www.fda.gov/food/nutrition-food-labeling-and-critical-foods/label-claims-food-dietary-supplements

Federal Trade Commission (2023, June 30). FTC sends cease and desist letters with FDA to companies selling edible products containing delta-8 THC in packaging nearly identical to food children eat. Retrieved March 3, 2025, from https://www.ftc.gov/news-events/news/press-releases/2023/07/ftc-sends-cease-desist-letters-fda-companies-selling-edible-products-containing-delta-8-thc

Francis, M. (2024, August 19). 'Gas-station heroin' targeted in crackdowns by states and cities. *NBC News*. Retrieved March 31, 2025, from https://www.yahoo.com/news/gas-station-heroin-targeted-crackdowns-150000486.html

Gala, K. (n.d.). Retrieved February 10, 2025, from https://kratomgala.com/

Global Kratom Coalition (2024). KCAC Position statement: kratom use in children. Retrieved March 3, 2025, from https://static1.squarespace.com/static/6508b3f79033221c2aa1ea17/t/677c4aed329dad0ce52dfd8a/1736198893809/KCAC+Positioning+Doc_1+Final.pdf

Global Kratom Coalition (2025, May 22). Regulatory map. https://globalkratomcoalition.org/regulation-map

Great CBD Shop (n.d.). Retrieved February 10, 2025, from https://greatcbdshop.com/

Heflin, J. O. (2023). Kratom regulation: federal status and state approaches. Congressional Research Service. Retrieved March 3, 2025, from https://crsreports.congress.gov/product/pdf/LSB/LSB11082

Hill, R., Kruegel, A. C., Javitch, J. A., Lane, J. R., & Canals, M. (2022). The respiratory depressant effects of mitragynine are limited by its conversion to 7-OH mitragynine. *British Journal of Pharmacology*, *179*(14), 3875–3885. https://doi.org/10.1111/bph.15832

Kruegel, A. C., Uprety, R., Grinnell, S. G., Langreck, C., Pekarskaya, E. A., Le Rouzic, V., Ansonoff, M., Gassaway, M. M., Pintar, J. E., Pasternak, G. W., Javitch, J. A., Majumdar, S., & Sames, D. (2019). 7-Hydroxymitragynine is an active metabolite of mitragynine and a key mediator of its analgesic effects. *ACS Central Science*, *5*(6), 992–1001. https://doi.org/10.1021/acscentsci.9b00141

McCurdy, C. R., Sharma, A., Smith, K. E., Veltri, C. A., Weiss, S. T., White, C. M., & Grundmann, O. (2024). An update on the clinical pharmacology of kratom: Uses, abuse potential, and future considerations. *Expert Review of Clinical Pharmacology*, *17*(2), 131–142. https://doi.org/10.1080/17512433.2024.2305798

Oberlo (2024). US search engine market share. Retrieved May 22, 2025, from https://www.oberlo.com/statistics/us-search-engine-market-share

OPMS (n.d.). Retrieved February 10, 2025, from https://buyopms.com/

Ponglux, D., Wongseripipatana, S., Takayama, H., Kikuchi, M., Kurihara, M., Kitajima, M., Aimi, N., & Sakai, S. (1994). A new indole alkaloid, 7α-hydroxy-7*H*-mitragynine, from Mitragyna speciosa in Thailand. *Planta Medica*, *60*(6), 580–581. https://doi.org/10.1055/s-2006-959578

Pure Leaf Kratom (n.d.). Retrieved February 10, 2025, from https://pureleafkratom.com/

Smith, K. E., Boyer, E. W., Grundmann, O., McCurdy, C. R., & Sharma, A. (2025). The rise of novel, semi-synthetic 7-hydroxymitragynine products. *Addiction*, *120*(2), 387–388. https://doi.org/10.1111/add.16728

Smith, K. E., Feldman, J. D., Freinberg, C., Shade, S., & Grundmann, O. (2025). Systematic review of clinical case reports involving tianeptine products: "Gas station heroin" hype or fact? *Current Addiction Reports*, *12*(1), 31. https://doi.org/10.1007/s40429-025-00640-z

Soneji, S. S., Knutzen, K. E., & Villanti, A. C. (2019). Use of flavored e-cigarettes among adolescents, young adults, and older adults: Findings from the Population Assessment for Tobacco and Health Study. *Public Health Reports*, *134*(3), 282–292. https://doi.org/10.1177/0033354919830967

Takayama, H., Kurihara, M., Kitajima, M., Said, I. M., & Aimi, N. (1998). New indole alkaloids from the leaves of Malaysian Mitragyna speciosa. *Tetrahedron*, *54*(29), 8433–8440. https://doi.org/10.1016/S0040-4020(98)00464-5

The Network for Public Health Law (2022, October 25). Cannabis regulation fact sheet. Retrieved March 3, 2025, from https://www.networkforphl.org/wp-content/uploads/2022/11/Packaging-Regulation.pdf

The Powder Solutions (n.d.). Retrieved February 10, 2025, from https://thepowdersolutions.com/

Váradi, A., Marrone, G. F., Palmer, T. C., Narayan, A., Szabó, M. R., Le Rouzic, V., Grinnell, S. G., Subrath, J. J., Warner, E., Kalra, S., Hunkele, A., Pagirsky, J., Eans, S. O., Medina, J. M., Xu, J., Pan, Y.-X., Borics, A., Pasternak, G. W., McLaughlin, J. P., & Majumdar, S. (2016). Mitragynine/corynantheidine pseudoindoxyls as opioid analgesics with mu agonism and delta antagonism, which do not recruit beta-arrestin-2. *Journal of Medicinal Chemistry*, *59*(18), 8381–8397. https://doi.org/10.1021/acs.jmedchem.6b00748

White, C. M. (2024a). Kratom's use and impact on pediatric populations. *Journal of Pediatric Pharmacology and Therapeutics*, 30(2),289-293. https://doi.org/10.5863/1551-6776-00.0.1

White, C. M. (2024b, February 6). 'Gas station heroin' is a dangerous and often contaminated supplement. *The Conversation*. Retrieved March 31, 2025, from https://www.scientificamerican.com/article/gas-station-heroin-is-a-dangerous-and-often-contaminated-supplement/

Wilson, L. L., Chakraborty, S., Eans, S. O., Cirino, T. J., Stacy, H. M., Simons, C. A., Uprety, R., Majumdar, S., & McLaughlin, J. P. (2021). Kratom alkaloids, natural and semi-synthetic, show less physical dependence and ameliorate opioid withdrawal. *Cellular and Molecular Neurobiology*, *41*(5), 1131–1143. https://doi.org/10.1007/s10571-020-01034-7

Yamamoto, L. T., Horie, S., Takayama, H., Aimi, N., Sakai, S., Yano, S., Shan, J., Pang, P. K., Ponglux, D., & Watanabe, K. (1999). Opioid receptor agonistic characteristics of mitragynine pseudoindoxyl in comparison with mitragynine derived from Thai medicinal plant Mitragyna speciosa. *General Pharmacology*, *33*(1), 73–81. https://doi.org/10.1016/s0306-3623(98)00265-1

Zuarth-Gonzalez, J. D., Ragsdale, A. K., Mukhopadhyay, S., McCurdy, C.R., McMahon, L.R., Wilkerson, J.L. (2025). Mitragynine and 7-hydroxymitragynine: Bidirectional effects on breathing in rats. *Journal of Pharmacological and Toxicological Methods*, *133*. https://doi.org/10.1016/j.vascn.2025.107624